ASSESSMENT TOOL 1

COURSE CODE: CSA16 COURSE NAME: Data Warehousing and Data Mining
KAVINILAVU.S
192565038

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre-processing techniques like Data cleaning, Data integration, Data transformation and data Reduction, for effective data preparation (BL3,BL4)

Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.
Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	Claims per 100 Policyholders (yi)
3.2	15
4.0	18
5.5	25
4.8	20

SOLUTION:

Step 1 — Tabulate the data

i	Premium % (x_i)	Claims/100 (y_i)
1	3.2	15
2	4.0	18
3	5.5	25
4	4.8	20

Step 2 — Compute the means

Step 3 — Compute deviations and products

x_i	y_i	$(x_i-\bar{x})$	$(y_i-\bar{y})$	$(x_i-\bar{x})(y_i-\bar{y})$
3.2	15	−1.175	−4.5	5.2875
4.0	18	−0.375	−1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125
			Σ =	**12.25**

Step 4 — Apply the covariance formula

Answer: Cov (x, y) = **+3.0625** (a **positive** value).

Interpretation: Since the covariance is positive, premium increase rate and claim frequency move in the **same direction** — a **positive (direct) relationship**. As the premium increase % rises, the number of claims per 100 policyholders also tends to rise.

Q3. Equal-Width Binning — BMI Risk Categorization (5 Marks)

Q4. Data Reduction — Bin Medians for Medical Supply Costs (5 Marks)
Note: the given list wasn't actually in sorted order (it starts 120, 150, 160, 160, **140**… which dips down), so the first step is to sort it as the question intends.
Sorted data (27 values):
120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300
Step 1 — Partition into equal-depth bins (depth = 3, matching Q2's method)

Bin	Values	Sorted within bin	Median
1	120, 140, 145	120, 140, 145	**140**
2	145, 148, 150	145, 148, 150	**148**
3	150, 150, 155	150, 150, 155	**150**
4	155, 155, 155	155, 155, 155	**155**
5	160, 160, 165	160, 160, 165	**160**
6	170, 170, 175	170, 170, 175	**170**
7	175, 175, 175	175, 175, 175	**175**
8	180, 190, 200	180, 190, 200	**190**
9	210, 230, 300	210, 230, 300	**230**

Step 2 — Replace every value in a bin with the bin's median

Bin	Original values	Reduced (bin median)
1	120, 140, 145	140, 140, 140
2	145, 148, 150	148, 148, 148
3	150, 150, 155	150, 150, 150
4	155, 155, 155	155, 155, 155
5	160, 160, 165	160, 160, 160
6	170, 170, 175	170, 170, 170
7	175, 175, 175	175, 175, 175
8	180, 190, 200	190, 190, 190
9	210, 230, 300	230, 230, 230

Result: the 27 individual cost records are reduced to just **9 representative values** (140, 148, 150, 155, 160, 170, 175, 190, 230) — this is bin-median-based data reduction, and it's noticeably more robust to the extreme outlier (300) than bin-mean smoothing would be, since 300 only pulls its own bin's median (230) mildly rather than distorting a mean.

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analysing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):
110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

Data (27 values, already sorted):
110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180
Assumption: bin depth = 3 (each bin holds 3 values → 9 bins), the standard equi-depth setup for this method.

Step 1 — Partition into bins of equal depth :

Bin	Values	Sum	Bin Mean
1	110, 112, 115	337	**112.33**
2	115, 118, 120	353	**117.67**
3	120, 122, 124	366	**122.00**
4	124, 126, 126	376	**125.33**
5	126, 126, 130	382	**127.33**
6	135, 135, 138	408	**136.00**
7	138, 138, 138	414	**138.00**
8	140, 145, 150	435	**145.00**
9	152, 160, 180	492	**164.00**

Step 2 — Smooth by bin means

Original	Smoothed	Original	Smoothed	Original	Smoothed
110	112.33	124	125.33	138	138.00
112	112.33	126	127.33	138	138.00
115	112.33	126	127.33	140	145.00
115	117.67	130	127.33	145	145.00
118	117.67	135	136.00	150	145.00
120	117.67	135	136.00	152	164.00
120	122.00	138	136.00	160	164.00

122	122.00	138	138.00	180	164.00
124	122.00	138	138.00		

Result: The sharp fluctuations (e.g., 152 → 160 → 180) are flattened into smoother, representative values, which reduces noise while preserving the overall trend.

3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.
To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

SOLUTION:

Data (18 values, sorted):
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5
Assumption: 4 equal-width bins (mirrors the 4 standard BMI risk categories).
Step 1 — Compute bin width

Step 2 — Define bin boundaries

Bin	**Range**
Bin 1	18.50 – 23.75
Bin 2	23.75 – 29.00
Bin 3	29.00 – 34.25
Bin 4	34.25 – 39.50

Step 3 — Assign each value and smooth by *nearest boundary*

Value	**Bin**	**Distance to lower / upper boundary**	**Replaced with**
18.5	1	0.00 / 5.25	**18.50**
19.2	1	0.70 / 4.55	**18.50**
21.8	1	3.30 / 1.95	**23.75**
24.9	2	1.15 / 4.10	**23.75**
25.5	2	1.75 / 3.50	**23.75**
26.2	2	2.45 / 2.80	**23.75**
26.4	2	2.65 / 2.60	**29.00**
27.8	2	4.05 / 1.20	**29.00**
29.2	3	0.20 / 5.05	**29.00**

30.2	3	1.20 / 4.05	**29.00**
30.4	3	1.40 / 3.85	**29.00**
31.9	3	2.90 / 2.35	**34.25**
32.4	3	3.40 / 1.85	**34.25**
33.1	3	4.10 / 1.15	**34.25**
33.6	3	4.60 / 0.65	**34.25**
34.7	4	0.45 / 4.80	**34.25**
38.2	4	3.95 / 1.30	**39.50**
39.5	4	0.00 / 5.25	**39.50**

Result: all 18 BMI values are reduced to just 4 distinct boundary values (18.50, 23.75, 29.00, 34.25, 39.50), simplifying risk classification while retaining the low/normal/overweight/obese structure.

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**.
To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):
120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

SOLUTION:

Sorted data (27 values):
120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300
Step 1 — Partition into equal-depth bins (depth = 3, matching Q2's method)

Bin	Values	Sorted within bin	Median
1	120, 140, 145	120, 140, 145	**140**
2	145, 148, 150	145, 148, 150	**148**
3	150, 150, 155	150, 150, 155	**150**
4	155, 155, 155	155, 155, 155	**155**
5	160, 160, 165	160, 160, 165	**160**
6	170, 170, 175	170, 170, 175	**170**
7	175, 175, 175	175, 175, 175	**175**
8	180, 190, 200	180, 190, 200	**190**
9	210, 230, 300	210, 230, 300	**230**

Step 2 — Replace every value in a bin with the bin's median

Bin	Original values	Reduced (bin median)

1	120, 140, 145	140, 140, 140
2	145, 148, 150	148, 148, 148
3	150, 150, 155	150, 150, 150
4	155, 155, 155	155, 155, 155
5	160, 160, 165	160, 160, 160
6	170, 170, 175	170, 170, 170
7	175, 175, 175	175, 175, 175
8	180, 190, 200	190, 190, 190
9	210, 230, 300	230, 230, 230

Result: the 27 individual cost records are reduced to just **9 representative values** (140, 148, 150, 155, 160, 170, 175, 190, 230) — this is bin-median-based data reduction, and it's noticeably more robust to the extreme outlier (300) than bin-mean smoothing would be, since 300 only pulls its own bin's median (230) mildly rather than distorting a mean

---

5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.

SOLUTION:

Data: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28 (n = 10)
Step 1 — Compute the mean

(a) Mean Absolute Deviation (MAD)

x_i	$\lvert x_i - 33\rvert$
18	15
22	11
25	8
42	9
28	5
43	10

33	0
35	2
56	23
28	5
Σ =	**88**

MAD = 8.8

(b) Z-scores for the first four observations

Using the MAD-based standardization formula (robust z-score):

x_i	**$x_i - 33$**	**$z_i = (x_i - 33)/8.8$**
18	−15	**−1.7045**
22	−11	**−1.2500**
25	−8	**−0.9091**
42	+9	**+1.0227**

Interpretation: Negative z-scores (18, 22, 25) mean those ages are below the group mean of 33; the positive z-score (42) means that age is above the mean, by just over one MAD.